



**INSTITUTE OF ENGINEERING AND TECHNOLOGY, KANNAD
DIST.CHH.SAMBHAJINAGAR**

A

PROJECT REPORT

On

**“IOT BASED INDUSTRIAL FAULT DETECTION AND SAFETY
SYSTEM”**

Submitted by

Mr. KALWANE GOKUL KRUSHNA

Ms. BODKHE PRIYANKA DNYANESHWAR

Ms. PAWAR POOJA BALU

Ms. DHONE NIKITA SURESH

In partial fulfillment for the award

Of

DIPLOMA

In

ELECTRONICS & TELECOMMUNICATION ENGG.

UNDER THE GUIDENCE OF

PROF. HAJARE.P.G

FOR THE ACADEMIC YEAR 2023-24

**MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION
[MSBTE]**



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**MAHARASHTRA STATE BOARD OF TECHNICAL
EDUCATION
CERTIFICATE**

This is to certify that,

Mr. KALWANE GOKUL KRUSHNA (2211570335)

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Of

Sixth Semester of

Diploma in Electronics and Telecommunication have completed the
Project entitled **“IOT BASED INDUSTRIAL FAULT DETECTION
AND SAFETY SYSTEM”**

Satisfactorily for the academic year 2023 to 2024 as prescribed in the curriculum of
MSBTE at **Institute of Engineering and Technology (polytechnic), kannad.**

Place: kannad

Date: 15/04/2024

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Mr. KALWANE GOKUL KRUSHNA

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ABSTRACT

IoT based industrial fault detection and diagnosis system. The IoT (Internet of things) based industry protection system employing Arduino is intended to safeguard industries from losses brought on by accidents and monitoring the faults. Industrial accidents such as gas leakage causes fire resulting in significant industrial setbacks. Due to furnace explosions, electrical short circuits, or other circumstances, quick fire detection is also required. The proposed system uses flame sensor along with gas sensor to detect fire as well as gas leakage to avoid any industrial mishaps and prevent economic damage. The system consists of temperature detector to detect the high/low temperature, the phase fault detector (either single or three phases), voltage controller (high/low), and frequency monitoring. Flame and Gas sensors are also interfaced with arduino and LCD screen. The sensor data is constantly scanned to record values and check for fire, gas leakage and then this data is transmitted to online. GSM module is sent message to specific numbers in case human operators aren't present in online always.

STUDENT DECLARATION

I hereby declare that my project report titled **IOT BASED INDUSTRIAL FAULT DETECTION AND SAFETY SYSTEM** is a bonafide record of the project work which I have submitted to institute of engineering and technology, kannad, in partial fulfillment of the credit requirements for the degree of diploma,

Mr. KALWANE GOKUL KRUSHNA

Ms. BODKHE PRIYANKA DNYANESHWAR

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1.INTRODUCTION

Now a days in industry lots of accident occur. There is a considerable potential of fire hazards in industries like petroleum, chemicals, oil and gas, which might cause enormous destruction, loss of property, and most importantly, loss of life. It is crucial to have a mechanism in place that can keep the area secure and alert the relevant stakeholders with in the allotted time if such an incident occurs. In our country also these types of accident are occurred. This project “Design and Implementation of Industry protection and fault monitoring system” can be reduces the accident and reduces the losses. The Internet of Things (IoT) industry protection system incorporating Arduino is a system created to guard against losses from incidents involving IoT. The system comprises of temperature, gas, and fire sensors interconnected to an Arduino and an LCD display. In order to capture readings and check for fire, gas leaks, or low/high temperatures, the sensor data is continuously scanned. This data is then provided online. In order to access the functions of internet, the gsm module is used. By using GSM module if the authority persons are not available in online it will send a SMS of their mobile phone. So that they want to know about the accident. Here we have added the fault detection system where check & control the frequency, phase and voltage. Power supply networks are constantly expanding, and their dependability is more crucial than ever. The intricacy of the entire network consists of several components that are prone to phase failure and can disrupt the electricity supply to end users. For the majority of the world's operating industries, low voltage, high voltage, and low frequency all exist.

1.1 Necessity: - If any of these sensors exceeds the threshold level, a buzzer is activated and wireless technology is used to send a warning notification to a wireless communication medium.

1.2 Objective of the project: - The prime objective of this study was to make sure the industry is accident free and reduce the loss of poverty and human life as well as to provide a healthy environment for the workers to work in. For that, the study also aimed to make the industry protection system wireless and to record defects and report observations and actions taken in relation to a specific defect to anticipated field engineers in order to make sure the fault is adequately removed from the system.

1.3 Theme: - Automation, fault detection, iot, safety, protection.

2.LITERATURE REVIEW

Bhosale Kiran Uttam and Galande Abhijeet Baspusaheb (June- 2017) in their paper explained about the progress of IOT in Industrial Automation. The Internet of Things (IOT) is a network of tangible items or artifacts that are equipped with sensors, computer software and network connectivity to enable data collection and exchange. The emphasized that most significant criteria of residential and industrial security systems for individuals are safety against fire and raw gas leaks. Because different hazardous gases are processed in industries, it is crucial to focus on the security of the employees who work there. If a gas leak occurs, a standard security system notifies the employers by turning on a warning alarm. Within the internet network, this technology also makes it possible to take certain critical decisions from anywhere on the earth. Between the connected network and the network, Wi-Fi shield serves as a service point. ((Uttam et al., 2017)) Sagar Prem Lalwani,[1]

Mehakpreet Kaur Khurana, Swati Jaikumar Khandare, Obaid Ur Rehman Ansari and Dr. Sanjay B.Pokle brief about Monitoring and alerting system for industrial metrics based on IOT in their paper. Our world has become increasingly interlinked than we could have ever anticipated thanks to infrastructure technologies like the smart grid, smart houses, smart water networks, and intelligent transportation. The comprehensive understanding of these systems is typically based on the Internet of Things (IOT), whereby the entire physical infrastructure is closely associated to information and communication technologies through the use of sensors, and where intelligent monitoring and management can be accomplished through the use of networked embedded devices. To share various forms of data, these equipments will be connected to the internet.[2]

3.DESIGN METHODOLOGY

The project's design process is based on data, which is collected through several sorts of sensors like IR sensors, flame sensors, gas sensors, and temperature and humidity sensors. These sensors are installed in the required locations, such as where the gas has leaked. The temperature and humidity sensors are used to display the readings, the flame sensor is used to detect fire, and the IR sensor is used to notify unauthorized entry. These sensors gather data continuously and communicate it to the microcontroller (Arduino UNO). Arduino UNO is programmed with a specific threshold value. If the value is less than the threshold, the situation is normal. If it exceeds the threshold value, the Arduino UNO sends a signal to the corresponding output. If the gas is leaking, a buzzer sound is produced when the sense is very low. When the perception of smell is strong, the exhauster fan is activated. If a fire is detected by the flame sensor, the Arduino UNO is triggered, and the buzzer is on. Anyone who enters the industry without being allowed is tracked using a IR sensor. On the LCD panel, all data is shown 16*2. Over IoT, data is sent and shared utilizing a GSM module. From our GSM network, we can control the GSM module. The GSM module allows microcontrollers to connect to wireless networks. The Thing speak website is used to provide graphical representations of all types of data from sensors including gas, flame, IR, humidity, and temperature sensors. We have developed a fully autonomous IoT wireless sensor- actuator network for the monitoring of the environment.

INPUT	OUTPUT
FIRE SENSOR	BUZZER
GAS SENSOR	BUZZER
TEMPERTURE + HUMIDITY SENSOR	FAN
LDR SENSOR	LCD & LED
IR SENSOR	LED & BUZZER

4.RESOURCES USED

1. Controllers

- I. Arduino uno
- II. Transistor BD139

2. Sensors

- I. Film sensor
- II. DHT11 sensor
- III. LDR sensor
- IV. MQ2 sensor
- V. IR sensor
- VI. Solar panel

3. Output components

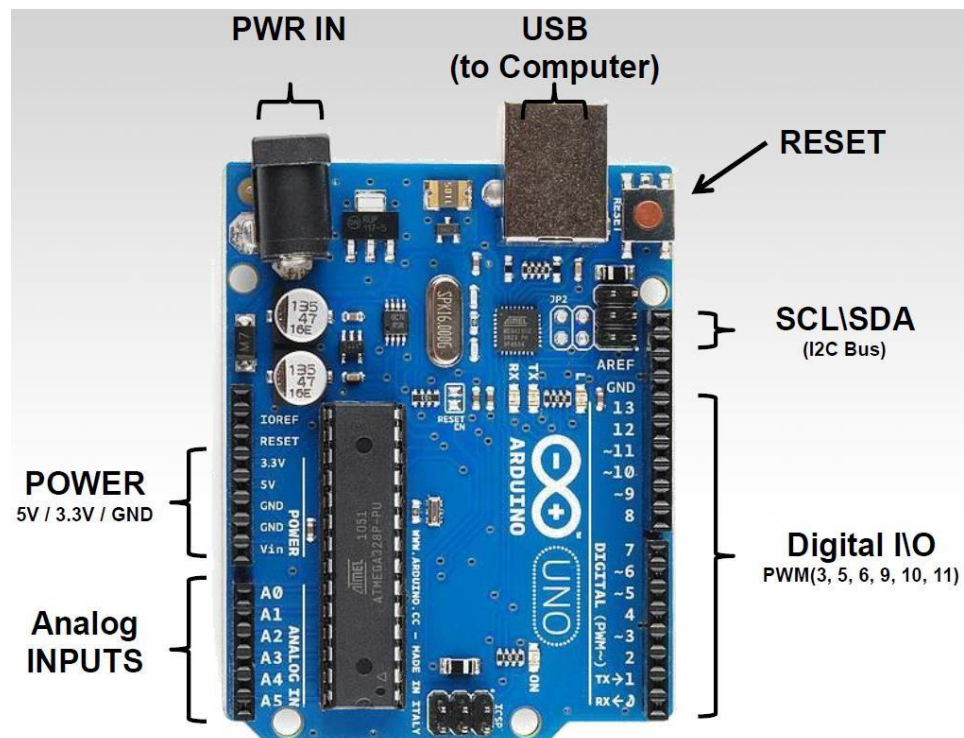
- I. Buzzer
- II. DC fan
- III. LED light
- IV. LCD display
- V. GSM module

5. Software used

- I. Arduino ide
- II. Blynk iot app & SMS

- **Components specification**

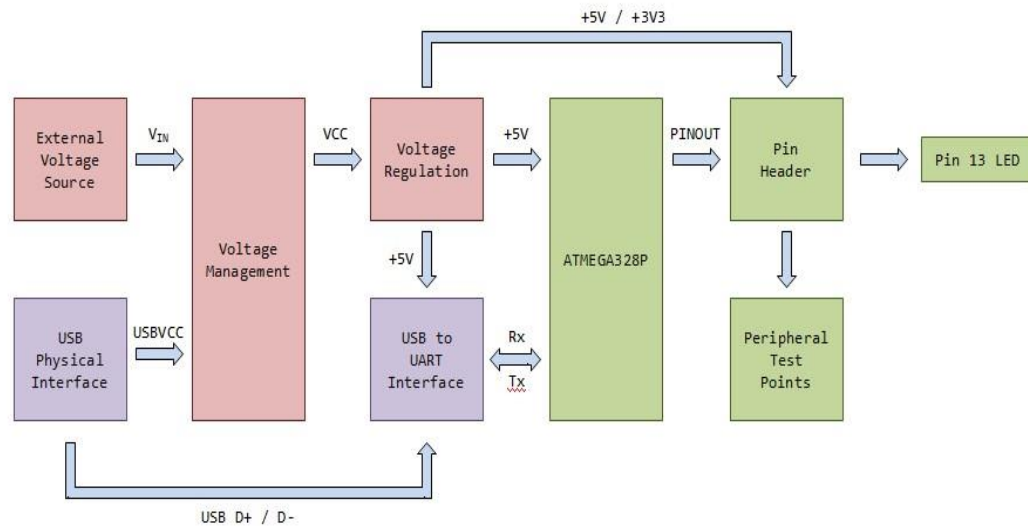
4.1 Arduino uno



Technical Specification

1. Microcontroller	ATmega328
2. Operating Voltage	5V
3. Input Voltage	7-12V
4. Input Voltage	6-20V
5. Digital I/O Pin	14 (of which 6 provide PWM)
6. Analog Input Pins	6
7. DC Current per I/O Pin	40 mA

8. DC Current for 3.3V Pin 50 mA
9. Flash Memory 32 KB
10. SRAM 2KB



11. EEPROM 1 KB
12. Clock Speed 16 MHz

Block diagram of arduino

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

4.2 GSM module (sim900A)

The SIM900A is a readily available GSM/GPRS module, used in many mobile phones and PDA. The module can also be used for developing IOT (Internet of Things) and Embedded Applications. SIM900A is a dual-band GSM/GPRS

engine that works on frequencies EGSM 900MHz and DCS 1800MHz. SIM900A features GPRS multi-slot class 10/ class 8 (optional) and supports the GPRS coding schemes CS-1, CS-2, CS-3 and CS-4.

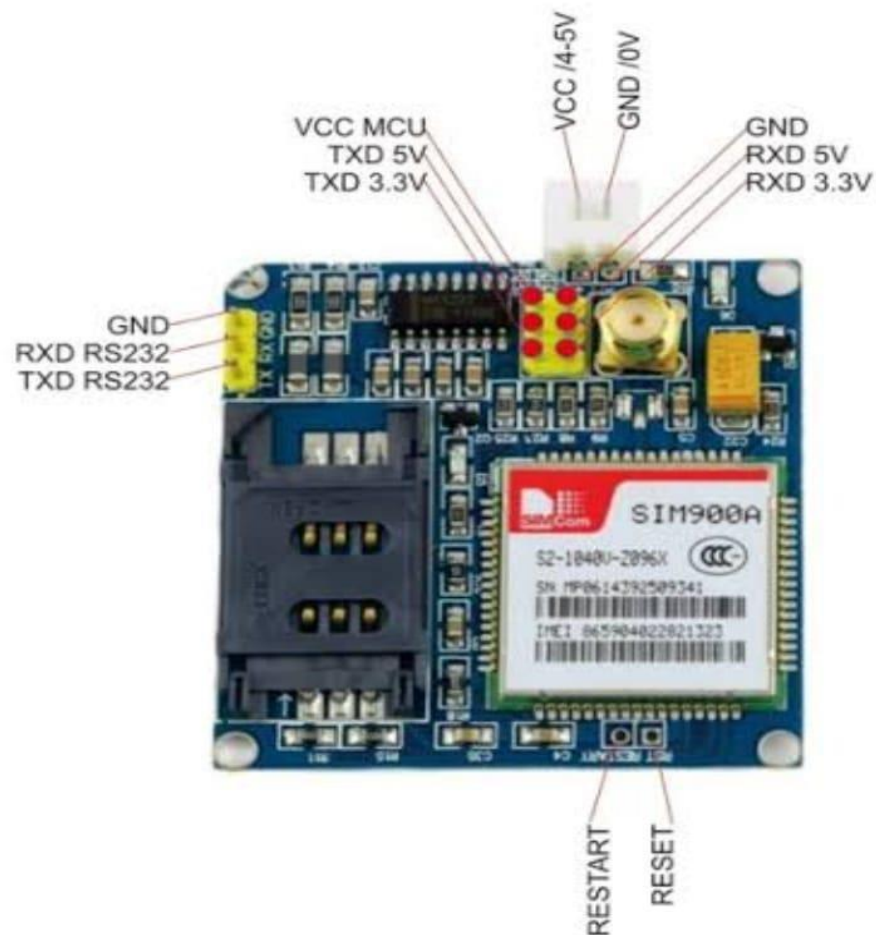


Fig.4.2 GSM module (sim900a)

- **Steps of connection for Arduino and GSM module interface**

We just need connect 4 wires to SIM900A module, that is power connection (VCC and GND). And Serial communication (RX-TX). Because we use arduino UNO which is

use 5V operating voltage and has 5V logic level (TTL). So, we need to connect arduino to 5RX and 5TX like in pinout picture above. Plug in sim card into sim card cartridge.

Features and Specifications

- Single supply voltage: 12v
- Power saving mode: Typical power consumption in SLEEP mode is 1.5mA
- Frequency bands: SIM900A Dual-band: EGSM900, DCS1800.
- GSM class: Small MS
- GPRS connectivity: GPRS multi-slot class 10 (default) , GPRS multi-slot class 8 (option)
- Transmitting power: Class 4 (2W) at EGSM 900, Class 1 (1W) at DCS 1800
- Operating Temperature: -30°C to +80°C
- Storage Temperature: -5°C to +90°C
- DATA GPRS: download transfer max is 85.6KBps, Upload transfer max 42.8KBps
- Supports CSD, USSD, SMS, FAX
- Supports MIC and Audio Input
- Speaker Input
- Features keypad interface
- Features display interface
- Features Real Time Clock
- Supports single SIM card
- Communication by using AT commands

4.3 MQ2 Sensors (gas & smoke detect)

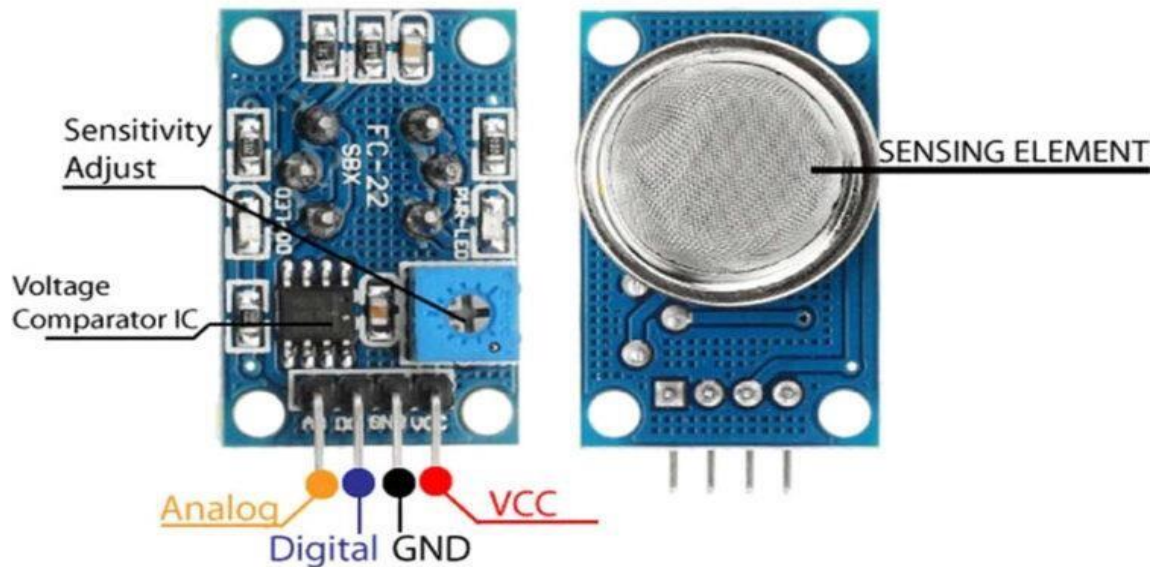


Fig4.3 MQ2 Sensors

Sensors are the electronic devices used for interaction with the outer environment. There are various types of sensors available that can detect light, noise, smoke, proximity etc. With the advent in technology, these are available as both analog and digital forms. Besides forming a communication with the outer environment, sensors are also a crucial part of safety systems. Fire sensors are used to detect the fire and take appropriate precautions on time. For smooth functioning of control systems and sensitive electronics, humidity sensors are used for maintaining humidity in the unit. One of such sensor used in safety systems to detect harmful gases is MQ2 Gas sensor.

• Features & specification

1. Operating Voltage is +5V
2. Can be used to Measure or detect LPG, Alcohol, Propane, Hydrogen, CO and even methane.
3. Analog output voltage: 0V to 5V.

4. Digital Output Voltage: 0V or 5V (TTL Logic).
5. Preheat duration 20 seconds.
6. Can be used as a Digital or analog sensor.
7. The Sensitivity of Digital pin can be varied using the potentiometer.

4.4 Flame Sensor (fire sensor)

Flame Sensor Module Infrared Receiver Ignition source detection module This tiny Flame sensor infrared receiver module ignition source detection module is Arduino compatible. It can use to detect flame or wavelength of the light source within 760nm~1100nm also useful for Lighter flame detection at a distance of 80cm. Greater the flame, the farther the test distance.

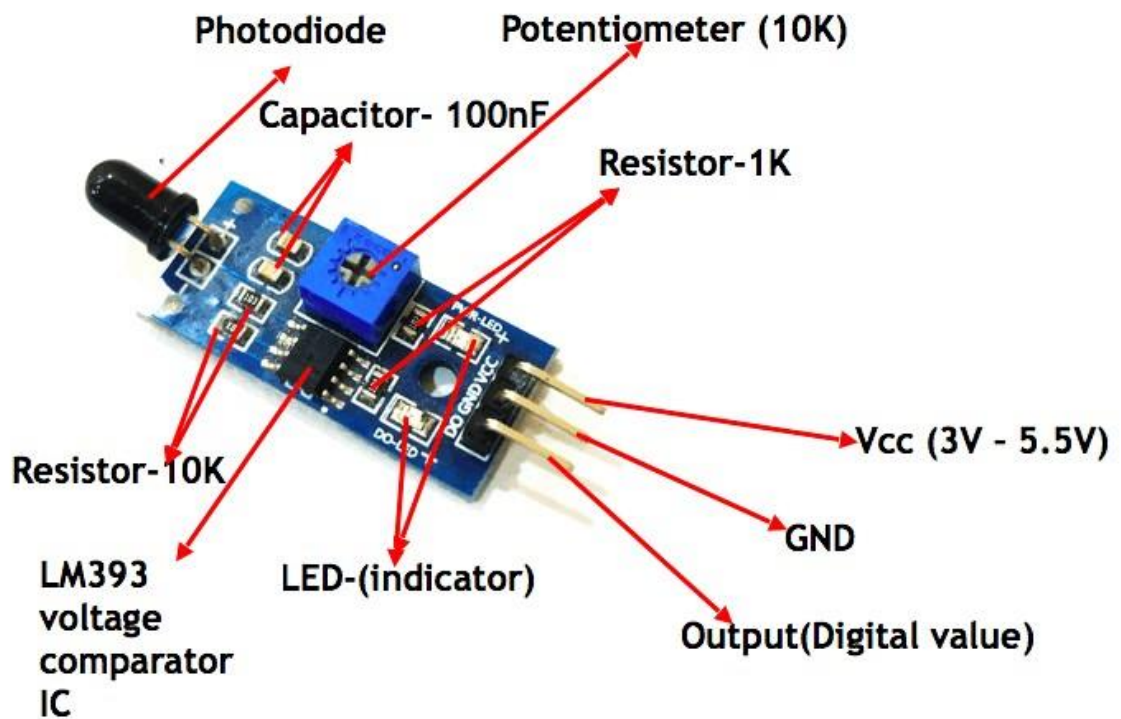


Fig4.4:- Flame sensor

It has a Detect angle of 60 degrees and is very sensitive to the flame spectrum. It produces the one-channel output signal at the D0 terminal for further processing like an alarm system or any switching system. The sensitivity is adjustable with the help of a blue potentiometer given on the board.

• **Features & specification**

1. Operating voltage: 3.3V to 5V
2. Operating current: 15 mA
3. Comparator chip used: LM393
4. Type of sensor: YG1006 Photo Transistor
5. Sensitivity can be adjusted by a potentiometer
6. Output type: Digital o/p or Digital & Analog output
7. Detection range: 60°
8. Detectable wavelength: 760 – 1100 nm
9. Working temperature: -25°C to 85°C

4.5 DHT11 Sensor

• **Working Principle of DHT11 Sensor**

DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature. The humidity sensing capacitor has two electrodes with a moisture holding substrate as a dielectric between them. Change in the capacitance value occurs with the change in humidity levels. The IC measure, process this changed resistance values and change them into digital form.

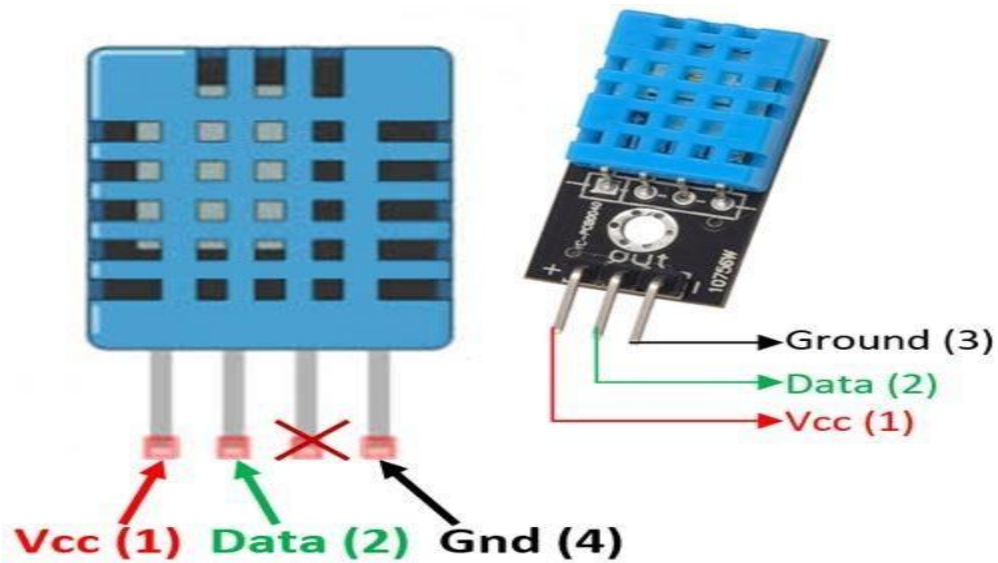


Fig4.5:- DHT11 sensor

For measuring temperature this sensor uses a Negative Temperature coefficient thermistor, which causes a decrease in its resistance value with increase in temperature. To get larger resistance value even for the smallest change in temperature, this sensor is usually made up of semiconductor ceramics or polymers.

The temperature range of DHT11 is from 0 to 50 degree Celsius with a 2degree accuracy. Humidity range of this sensor is from 20 to 80% with 5% accuracy. The sampling rate of this sensor is 1Hz .i.e. it gives one reading for every second. DHT11 is small in size with operating voltage from 3 to 5 volts. The maximum current used while measuring is 2.5mA.

• **DHT11 Specifications**

1. Operating Voltage: 3.5V to 5.5V
2. Operating current: 0.3mA (measuring) 60uA (standby)
3. Output: Serial data
4. Temperature Range: 0°C to 50°C
5. Humidity Range: 20% to 90%
6. Resolution: Temperature and Humidity both are 16-bit
7. Accuracy: $\pm 1^{\circ}\text{C}$ and $\pm 1\%$

4.6 IR Sensor: -

Working: - IR technology is used in a wide range of wireless applications which includes remote controls and sensing. The infrared part in the electromagnetic spectrum can be separated into three main regions: near IR, mid-IR & far IR. The wavelengths of these three regions vary based on the application. For the near IR region, the wavelength ranges from 700 nm- 1400 nm, the wavelength of the mid-IR region ranges from 1400 nm – 3000 nm & finally for the far IR region, the wavelength ranges from 3000 nm – 1 mm.

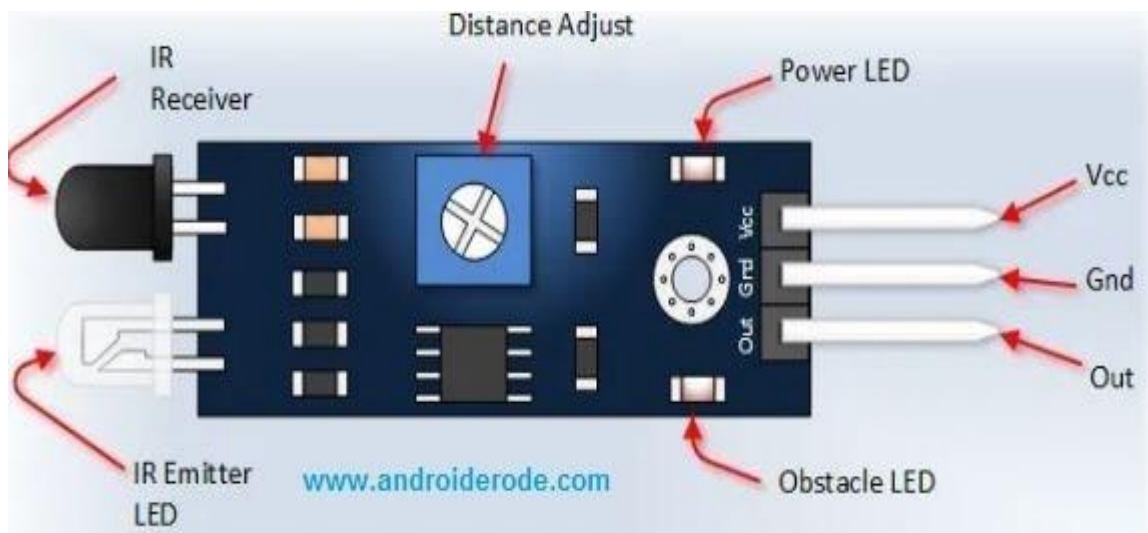


Fig4.6:- IR Sensor

The near IR region is used on fiber optic & IR sensors, the mid-IR region is used for heat sensing and the far IR region is used in thermal imaging. The range of frequency for IR is maximum as compared to microwave and minimum than visible light. This article discusses an overview of the IR sensor and its working.

specifications and features of the IR sensor

1. The operating voltage is 5VDC
2. I/O pins – 3.3V & 5V

3. Mounting hole
4. The range is up to 20 centimeters
5. The supply current is 20mA
6. The range of sensing is adjustable
7. Fixed ambient light sensor

4.7 LDR sensor: -

Working: -

First of all, we need to connect the LDR sensor module to a 5v power supply. Then set the threshold voltage at the Non-Inverting input (3) of the IC according to the present light intensity by rotating the preset knob for setting the sensor sensitivity.

When light intensity increase on the surface of the LDR then the resistance of the LDR decreases. Then the maximum amount of voltage will be allocated across the resistor(R3). So, a Low amount of voltage from the LDR is given to the Inverting input (2) of the IC. Then the Comparator IC compares this voltage with the threshold voltage. In this condition, this input voltage is less than the threshold voltage, so the sensor output goes LOW (0).



Fig. LDR Sensor

In contrast, When light intensity decrease (low/dark) on the surface of the LDR then the resistance of the LDR increases. Then the maximum amount of voltage will be allocated across the LDR (R2). So, a High amount of voltage from the LDR is given to the Inverting input (2) of the IC. Then the Comparator IC compares this voltage with the threshold voltage. In this condition, this input voltage is greater than the threshold voltage, so the sensor output goes High (1).

• Sensor Specifications

1. Operating voltage: 5V or 3.3V DC
2. Comparator chip: LM393
3. Module Pins: 3 pins
4. Output type: Digital outputs (D0)
5. Sensitivity: Adjustable
6. Indicator LED: Output and power LED indicator
7. PCB size : 3 cm * 1.6cm
8. Fixed Hole Diameter: 3mm

5.SYSTEM DEVELOPMENT

5.1 Block diagram

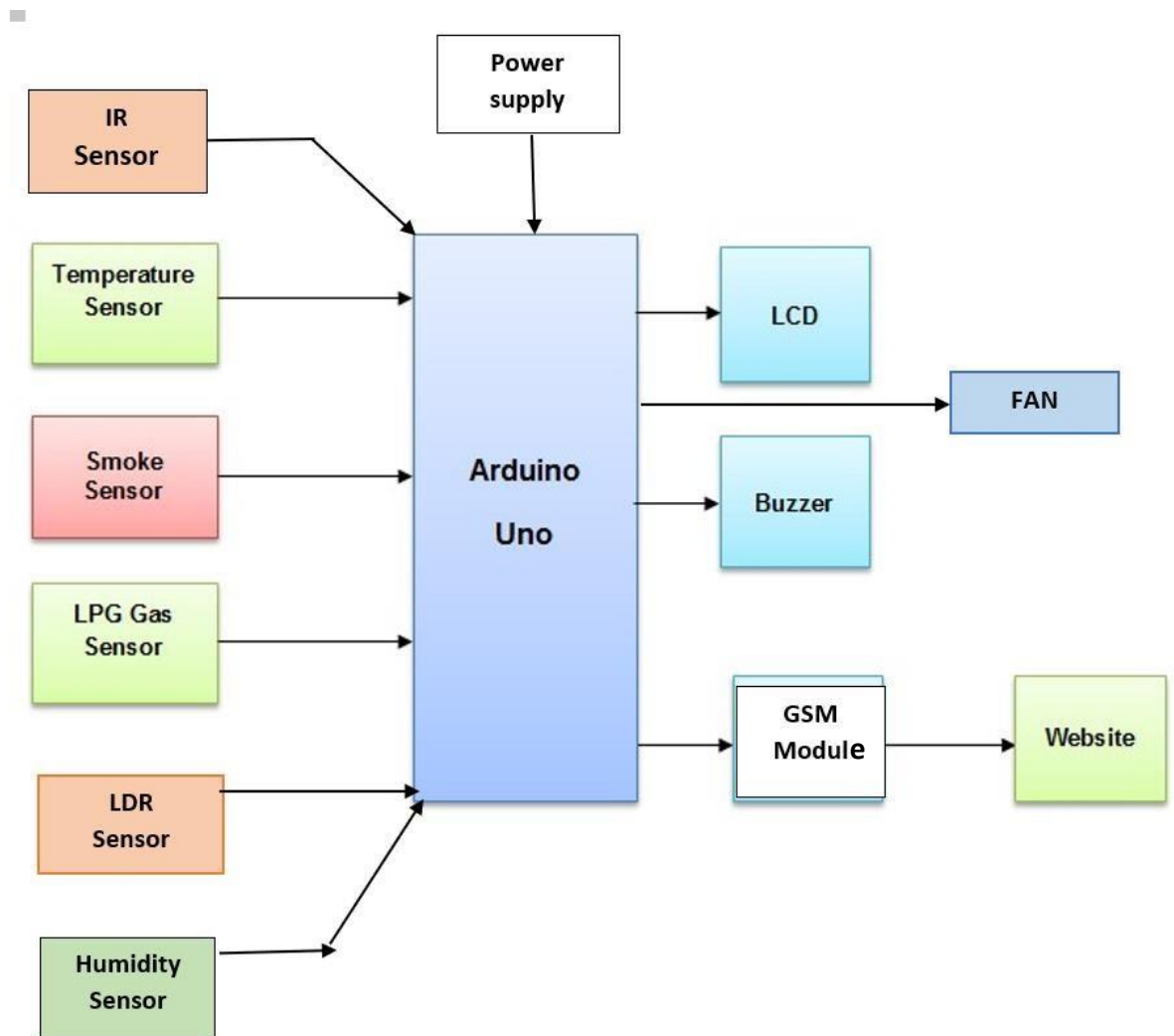


Fig 5.1: - block diagram

5.2 Circuit diagram

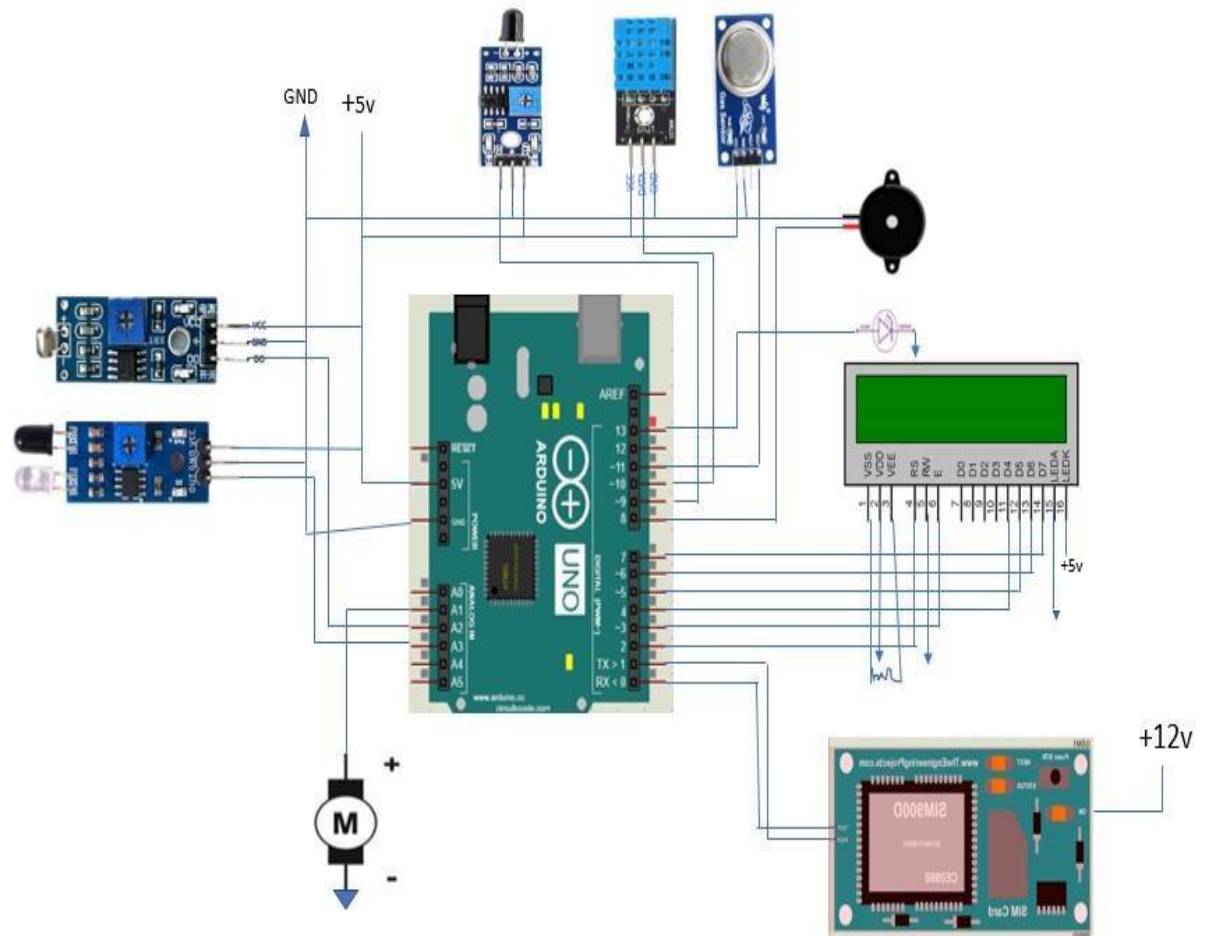


Fig 5.2: - circuit diagram

5.3 Working: -

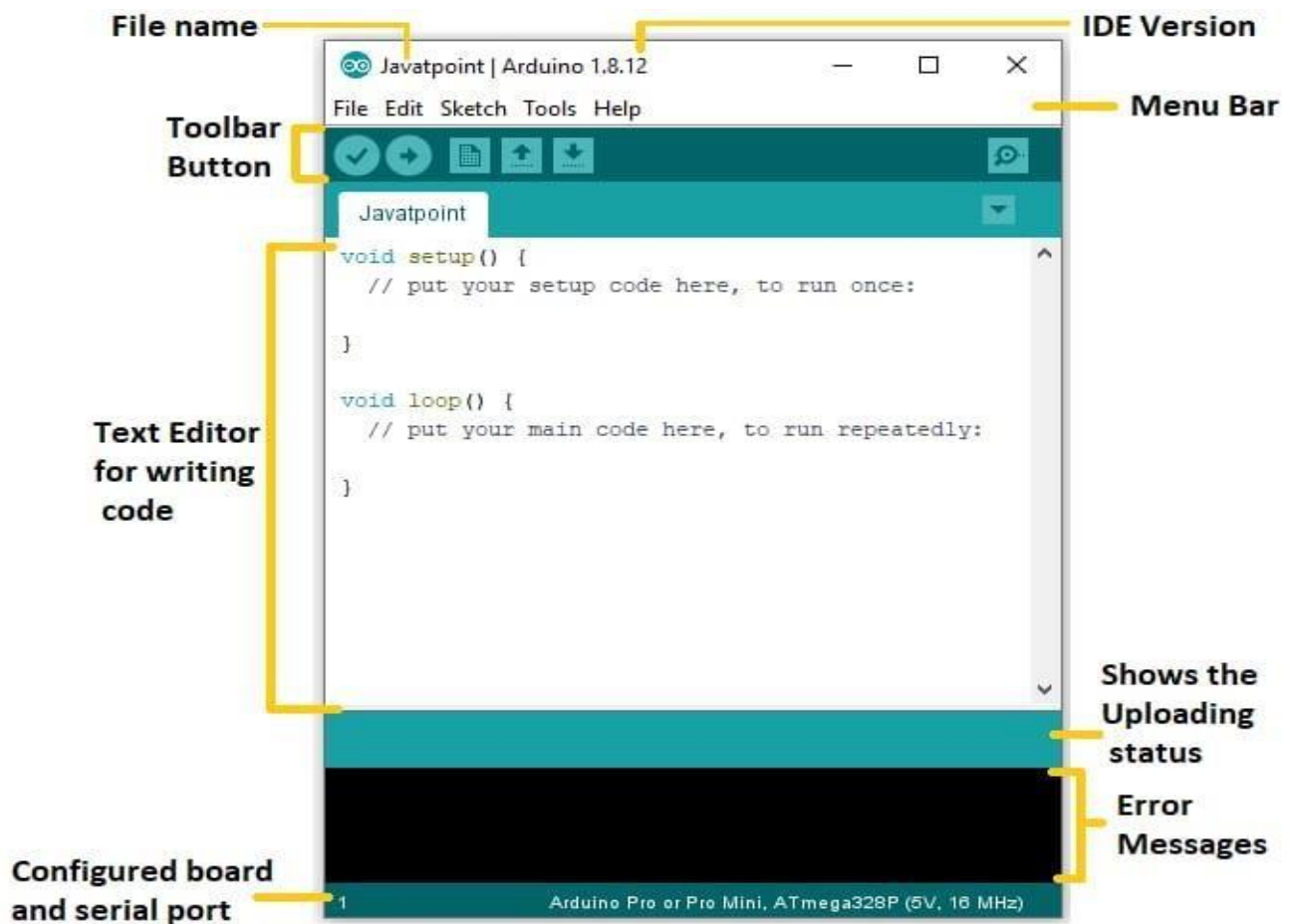
For designing the system uses Temperature sensor, Gas sensor, Fire sensor, humidity sensor, IR sensor, ldr sensor, ATmega328 controller, Power supply, android based mobile etc. this project detect the fault occurred in the industry and controls the industrial parameter. In the industry there is no immediate action taken by authorized person on the fault, so that in this project uses GSM module that will send the message instantly to respective authority. So that industrial place project is very useful. This system easily detect the fault when parameter exceed the threshold values. for controlling the parameter system uses DC fan, buzzer as a output devices. This devices are automatically turn on when any value goes above the threshold. In industry various faults are occurs because of the parameters goes above their range, this system uses DHT11 as a temperature and humidity sensor for detecting the temperature and humidity. This sensor will measure the room temperature of industry and if the temperature goes above a value this will be displayed on android application.

Ldr sensor detect the light and night time turn on the light. Application of IR sensor is object detect and used motion detect.

6. SOFTWARE DEVELOPMENT

6.1 Arduino ide: - The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. Here, IDE stands for Integrated Development Environment.

The program or code written in the Arduino IDE is often called as sketching. We need to connect the Genuino and Arduino board with the IDE to upload the sketch written in the Arduino IDE software.



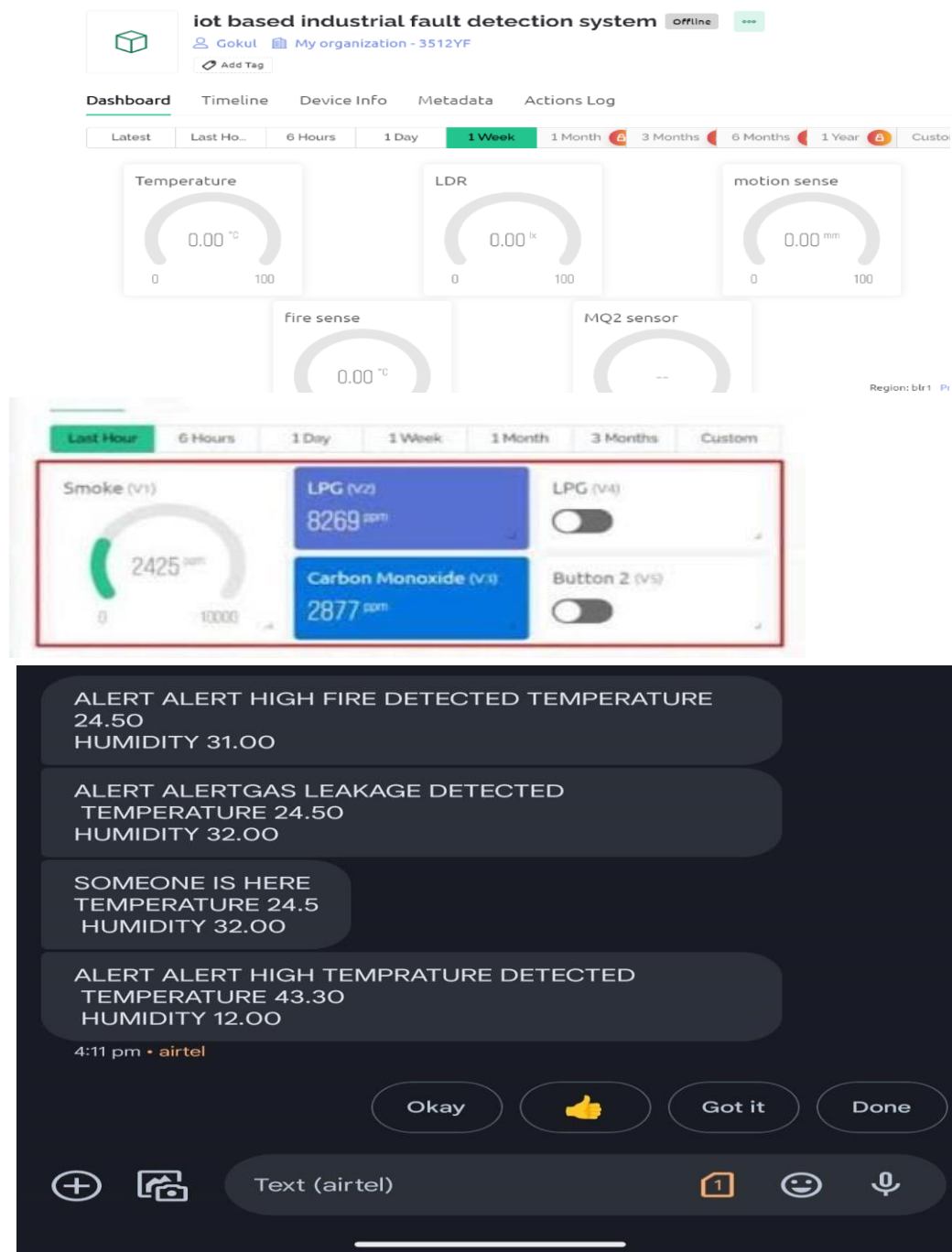
6.1.1 Concept of compiler: - Compiling a program is the process of transforming high level source code into a low-level object code (binary code) called machine language,

which can be understood by the processor. In Arduino IDE AVR-GCC Toolchain is used for compiling the program.



6.1.2 Concept of code upload: - In the Arduino environment, we write sketches that can be uploaded to Arduino boards. In this tutorial, we will go through how to select a board connected to your computer, and how to upload a sketch to that board, using the Arduino IDE 2.

6.2 Blynk iot application and SMS: - Blynk is a comprehensive software suite that enables the prototyping, deployment, and remote management of connected electronic devices at any scale. Whether it's personal IoT projects or commercial connected products in the millions, Blynk empowers users to connect their hardware to the cloud and create iOS, Android, and web applications, analyze real-time and historical data from devices, remotely control them from anywhere, receive important notifications, and much more.



6.2.1 Concept of cloud: - The cloud" refers to servers that are accessed over the Internet, and the software and databases that run on those servers. Cloud servers are located in data centers all over the world. By using cloud computing, users and companies do not have to manage physical servers themselves or run software applications on their own machines.

7.SOFTWARE CODE

```
#INCLUDE <LIQUIDCRYSTAL.H>

const int RS = 2, EN = 3, D4 = 4, D5 = 5, D6 = 6, D7 = 7;

LiquidCrystal lcd(RS, EN, D4, D5, D6, D7);

#include <DHT.h>

#define DHTPIN 10

#define DHTTYPE DHT11

DHT DHT(DHTPIN, DHTTYPE);

#define BUZZ 8

#define FLM 9

#define MQ 11

#define RLY A1

#define LDR A2

#define LED A0

#define IR A3

int FLMS;

int MQS;

int LDRS;

int IRS;

void setup()

{

    Serial.begin(9600);

    dht.begin();

    PinMode(LDR, INPUT);

    PinMode(FLM, INPUT);

    PinMode(BUZZ, OUTPUT);

    PinMode(MQ, INPUT);

    PinMode(IR, INPUT);
```

```

    PinMode(RLY, OUTPUT);
    PinMode(LED, OUTPUT);
    lcd.begin(16, 2);
    lcd.print("  WELCOME");
    delay(1000);
}

void loop()
{
    FLMS = digitalRead(FLM);
    MQS = digitalRead(MQ);
    LDRS = digitalRead(LDR);
    IRS = digitalRead(IR);
    FLOAT H = DHT.Readhumidity();
    FLOAT T = DHT.Readtemperature();
    if (ISNAN(H) || ISNAN(T)) {
        serial.println(F("FAILED TO READ FROM DHT SENSOR!"));
        RETURN;
    }
    if (FLMS == LOW)
    {
        digitalWrite(BUZZ, HIGH);
        lcd.clear();
        lcd.setCursor (0, 0);
        lcd.print("ALERT ALERT HIGH");
        lcd.setCursor(0, 1);
        lcd.print("FIRE DETECTED");
        ///////////////////////////////////
        delay(100);
        serial.println("AT+CMGF=1");  delay(1000);
        serial.println("AT+CMGF=1"); delay(1000);
    }
}

```

```

serial.println("AT+CMGF=1");
delay(1000);
serial.println("AT+CMGS=\"+919423746654\"");    delay(1000);
    serial.println("ALERT ALERT HIGH");
    delay(500);
    serial.println(" FIRE DETECTED");
    delay(500);
    serial.print(" TEMPERATURE ");
    serial.print(T);
    delay(500);
    serial.print(" HUMIDITY ");
    serial.println(H);
    delay(500);
    serial.Write(26);
    //////////////////////////////////////
    delay(3000);
}
if (T >= 40)
{
    digitalWrite(RLY, HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ALERT ALERT HIGH ");
    lcd.setCursor(0, 1);
    lcd.print("TEMPRATURE DETE...");
    //////////////////////////////////////
    delay(100);
    serial.println("AT+CMGF=1");    delay(1000);
    serial.println("AT+CMGF=1");    delay(1000);
    serial.println("AT+CMGF=1");
    delay(1000);

```

```

serial.println("AT+CMGS=\"+919423746654\"");
delay(1000);
    serial.println("ALERT ALERT HIGH");
    delay(500);
    serial.println(" TEMPRATURE DETECTED");
    delay(500);
    serial.println(" TEMPERATURE ");
    serial.println(T);
    delay(500);
    serial.println(" HUMIDITY ");
    serial.println(H);
    delay(500);
    serial.Write(26);
    //////////////////////////////////////
    delay(3000);
}
if (MQS == LOW)
{
    digitalWrite(BUZZ, HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("  ALERT ALERT");
    lcd.setCursor(0, 1);
    lcd.print("GAS LEAKAGE DETEC...");
    //////////////////////////////////////
    delay(100);
    serial.println("AT+CMGF=1");    delay(1000);
    serial.println("AT+CMGF=1");    delay(1000);
    serial.println("AT+CMGF=1");
    delay(1000);
    serial.println("AT+CMGS=\"+919423746654\"");

```

```

delay(1000);
serial.println("ALERT ALERT");
    delay(500);
    serial.println("GAS LEAKAGE DETECTED");
    delay(500);
    serial.println(" TEMPERATURE ");
    serial.println(T);
    delay(500);
    serial.println(" HUMIDITY ");
    serial.println(H);
    delay(500);
    serial.Write(26);
    //////////////////////////////////////
    delay(3000);
}
if (LDRS == LOW)
{
    digitalWrite(LED, HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("IT'S NIGHT TIME");
    lcd.setCursor(0, 1);
    lcd.print("TURN ON LIGHT");
    delay(3000);
}
if (IRS == HIGH)
{
    digitalWrite(LED, HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("SOMEONE IS HERE");
}

```

```

////////////////////////////////////
delay(100);
serial.println("AT+CMGF=1");
delay(1000);
    serial.println("AT+CMGF=1");    delay(1000);
    serial.println("AT+CMGF=1");
    delay(1000);
    serial.println("AT+CMGS=\"+919423746654\"");
    delay(1000);
    serial.println("SOMEONE IS HERE");
    delay(500);
    serial.println(" TEMPERATURE ");
    serial.println(T);
    delay(500);
    serial.println(" HUMIDITY ");
    serial.println(H);
    delay(500);
    serial.Write(26);
    //////////////////////////////////////
    delay(3000);
}
digitalWrite(BUZZ, LOW);
digitalWrite(RLY, LOW);
digitalWrite(LED, LOW);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("HUMI: ");
lcd.print(H);
lcd.print("%");
lcd.setCursor(0, 1);
lcd.print("TEMP: ");

```

```
lcd.print(T);  
lcd.print("*C ");  
delay(200);  
  
}
```

8.FUTURE SCOPE

- When there is gas leakage, the main wall is closed automatically.
- When shared the exact location when the fault was detected using gps module.
- When fire is detected, the fire extinguisher will be turned on automatically.
- Future Development of the project — We can add a circuit that can send out safety instructions like ‘vacate the place’, once fire OR Gas leakage is detected.

9.ADVANTAGES AND APPLICATION

I. Advantages

- 1) The probability of human error is greatly reduced.
- 2) This system provides time and financial savings for the business.
- 3) Constant monitoring also allows for consistent and dependable data from anywhere around the world.
- 4) Major accidents can be prevented without any huge damages.

II. Disadvantages

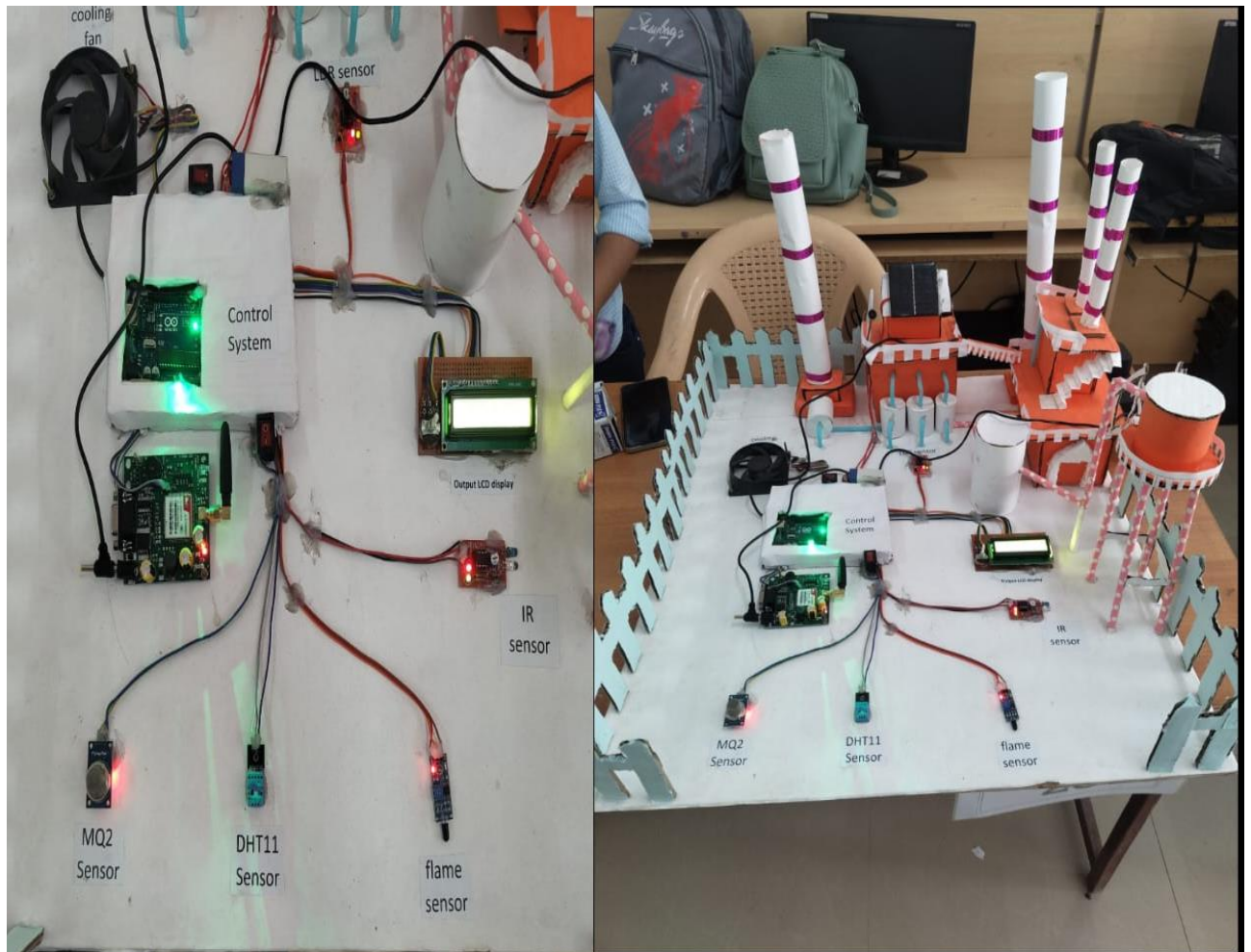
- 1) The complete range if the sensors is limited and so there is a need for multiple system
- 2) to exist for effective monitoring of the factory.

II. Applications

- 1) It is also used in various other industries like beverage manufacturing, biogas industry.
- 2) Laboratories contain many chemicals, in that many are retractable to high temperatures or heat.
- 3) Refrigeration plants use CFC gases which highly react able with fire, so it is necessary to have this system in the plants.
- 4) Safety purpose.
- 5) Home security and safety.

10.RESULT

After completing all implementation as per design, we turned power on then we confirmed that all the equipment works properly. The full mechanisms and tests are presented in the following sections. In the discussion, the result is analyzed and compared. As we have mentioned earlier that we have practically operated our device in different scenario. The final result was quite satisfactory. Now if we observe the data quality then we can see that the values from the sensors are quite accurate and to be more specific the sensors give us almost 100% accurate values.





11. CONCLUSION

By working on this project, we aim to gain in-depth knowledge in the technical aspects of the internet of things and digital electronics systems. The core objective of our industrial detection system is achieved with digital integration of sensors with optimum value precision and micro-controllers. We have incorporated sensors for detection of environmental parameters which when reaches undesired condition observed data is communicated to the head center. Arduino UNO is used for processing and allocation of the data received and the processed data is communicated to the internet applications via the internet. Therefore, we have included MQ2 gas sensor, flame sensor, IR sensor, temperature sensors for receiving data and Arduino for inter device communications and proposing desired commands. We have also used an internet-based Thing Speak website and SMS for the collection of data. The significance for preventing undesired hazards and activities in the management's absence is the key novelty of our project and we believe it would help industries reduce these accident rates efficiently if deployed. With the ongoing industrial modernization, these days the demand for digital and automation systems to perform functions especially in the information technology and algorithmic sectors has risen exponentially and the application of IoT in their operational mechanisms paves way for development and feasible innovation.

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